

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #24: Public Bus Live Tracking & Crowding Estimator
Jira Project: PBLT (PUBLIC BUS LIVE TRACKING)

Submission: Backlog, Epics, Sprints & Burndown Charts

Project Overview

This submission documents the Agile backlog and sprint simulation completed in Jira for the Public Bus Live Tracking & Crowding Estimator project (Problem Statement #24). The Lab 1 functional requirements were broken down into two Epics and six User Stories, prioritized, estimated using the Fibonacci scale, and executed across three simulated sprints.

Epics Created

Epic 1 (PBLT-1): Live Bus Tracking & ETA - Enable commuters to find buses, track their locations, and view accurate estimated arrival times.

Epic 2 (PBLT-5): Crowding, Fleet Monitoring & Alerts - Enable crowding visibility, fleet oversight, and delay notifications.

Epic 1: Live Bus Tracking & ETA (PBLT-1)

Description: Enable commuters to find buses, track their locations, and view accurate estimated arrival times.

Backlog & User Stories

The backlog below shows Epic 1 with its three User Stories - Search Stop & View ETA (5 pts), Real-Time Bus Location (5 pts), and Accurate ETA Updates (8 pts) - for a total estimate of 18 story points.

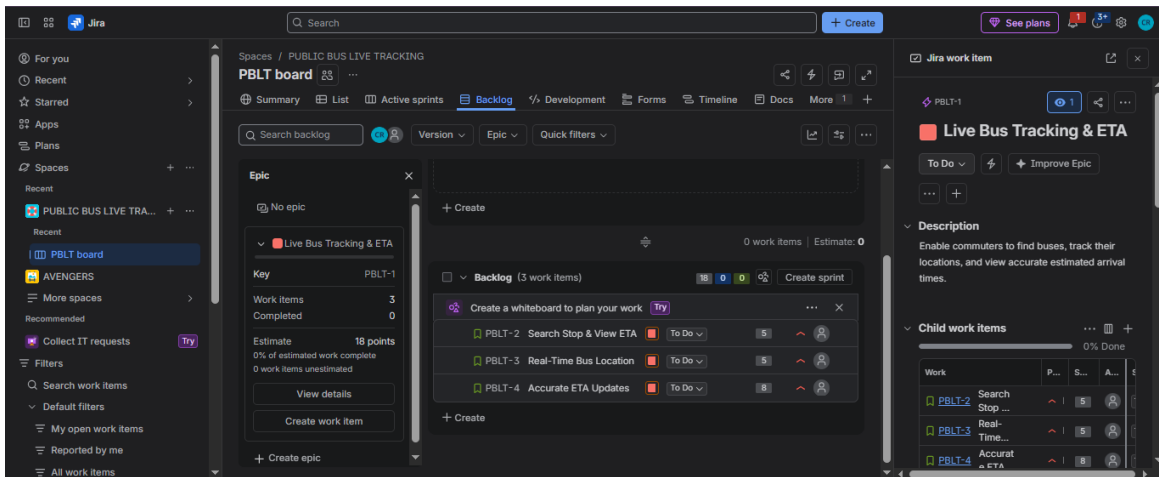


Fig 1.1 - Epic 1 backlog with child User Stories, priorities, and story points

Sprint 1 - Active Sprint Board

Sprint 1 included Real-Time Bus Location and Accurate ETA Updates in progress, with Search Stop & View ETA moved to Done.

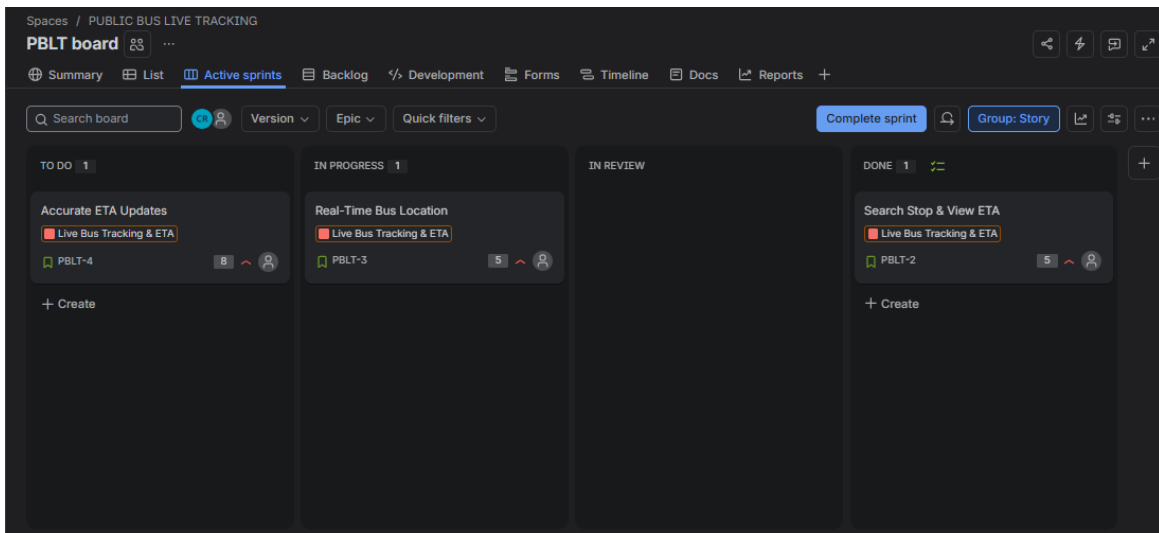


Fig 1.2 - Sprint 1 board (To Do / In Progress / Done)

Sprint 1 Burndown Chart

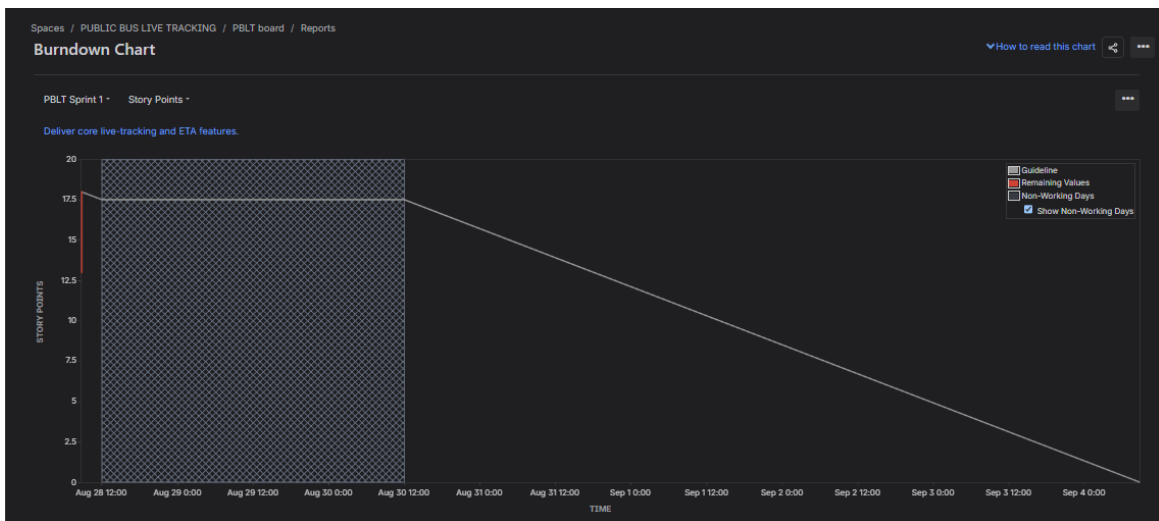


Fig 1.3 - Sprint 1 burndown chart (Goal: Deliver core live-tracking and ETA features)

Date	Issue	Event Type	Event Detail	Story Points		
				Int.	Dec.	Remaining
28/Aug/26 8:53 AM	PBLT-3	Sprint start		5		
	PBLT-4			8		
	Total					13
28/Aug/26 8:54 AM	PBLT-3	Burndown	Issue completed	5	8	
	PBLT-4	Burndown	Issue completed	8	0	
28/Aug/26 8:55 AM	PBLT-3	Sprint ended by Chathurth R				0
	PBLT-4					0
	Total					0

Fig 1.4 - Sprint 1 burndown event log

Sprint 2 - Backlog & Prioritization

The remaining Sprint 1 stories (Accurate ETA Updates - 8 pts, Real-Time Bus Location - 5 pts) were carried into Sprint 2, totaling 13 story points, while Search Stop & View ETA was completed.

Spaces / PUBLIC BUS LIVE TRACKING

PBLT board

Summary List Active sprints **Backlog** Development Forms Timeline Docs Reports +

Search backlog Version Epic Quick filters

Epic

No epic

> Live Bus Tracking & ETA

+ Create epic

PBLT Sprint 2 4 Sep - 11 Sep (2 work items) 8 5 0 Start sprint

☒ PBLT-4 Accurate ETA Updates Live Bus Tracking ... To Do 8

☒ PBLT-3 Real-Time Bus Location Live Bus Tracking ... In Progress 5

+ Create

2 work items | Estimate: 13

Backlog (1 work item) 0 0 5 Create sprint

☐ PBLT-2 Search Stop & View ETA Live Bus Tracking ... Done 5

+ Create

2 selected | Select all | Edit fields | Change status | Merge | Watch options | Delete | X

Fig 1.5 - Sprint 2 backlog selection with story points

Sprint 2 Burndown Chart

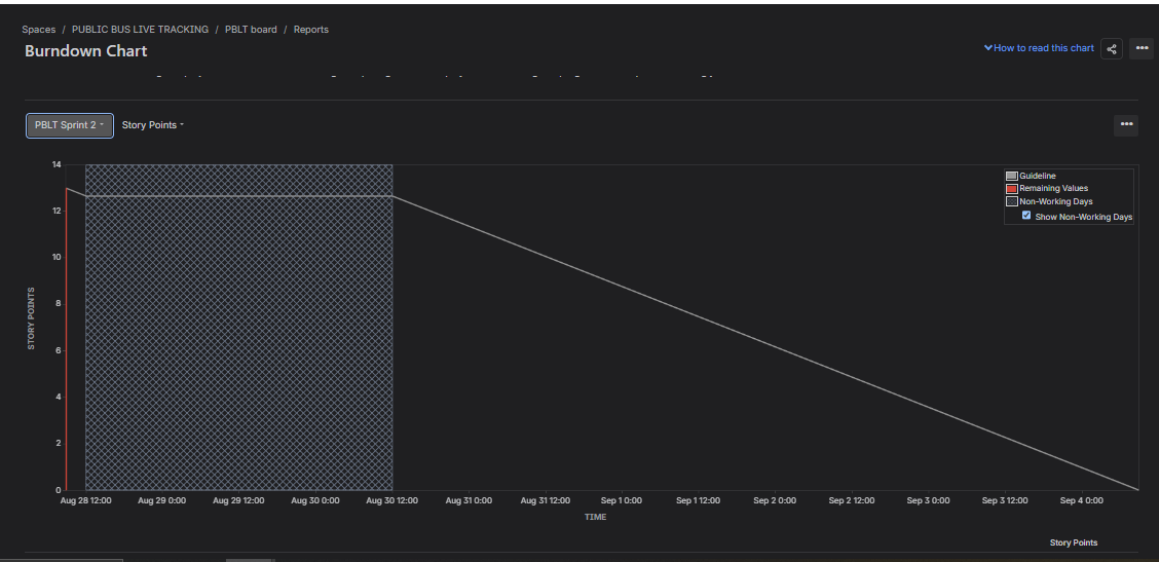


Fig 1.6 - Sprint 2 burndown chart

Date	Issue	Event Type	Event Detail	Story Points		
				Inc.	Dec.	Remaining
28/Aug/26 8:45 AM	PBLT-2	Sprint start		5		
	PBLT-3			5		
	PBLT-4			8		
	Total					18
28/Aug/26 8:47 AM	PBLT-3	Burndown	Issue completed		5	13
	PBLT-3	Burndown	Issue reopened	5		18
	PBLT-2	Burndown	Issue completed		5	13
28/Aug/26 8:52 AM	PBLT-2	Sprint ended by Chathurth R				0
	PBLT-3					5
	PBLT-4					8
	Total					13

Fig 1.7 - Sprint 2 burndown event log

Epic 2: Crowding, Fleet Monitoring & Alerts (PBLT-5)

Description: Provide crowding-level visibility to commuters, fleet status monitoring, and delay alerts.

Backlog & Story Points

Epic 2 contains three User Stories - View Crowding Level (5 pts), Monitor Fleet Status (8 pts), and Receive Delay Alert (5 pts).

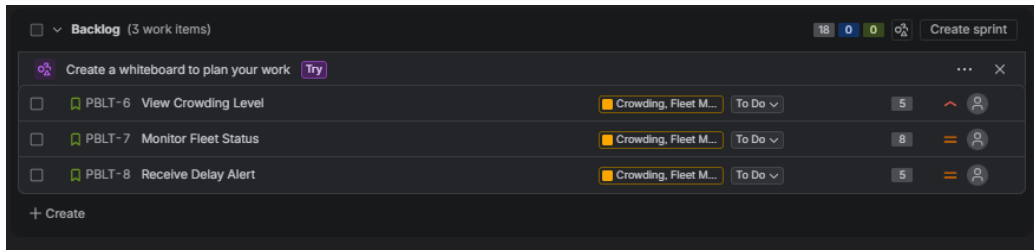


Fig 2.1 - Epic 2 backlog with story points assigned

Epic Details Panel

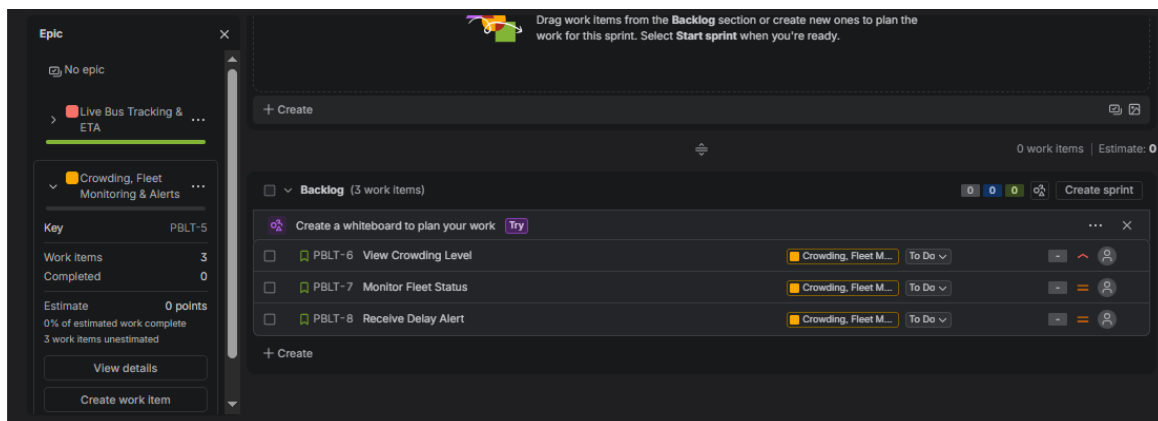


Fig 2.2 - Epic 2 details panel showing associated backlog items

Sprint 3 - Active Sprint Board

Sprint 3 combined all three Epic 2 stories. View Crowding Level and Monitor Fleet Status were completed, while Receive Delay Alert remained In Progress at sprint close.

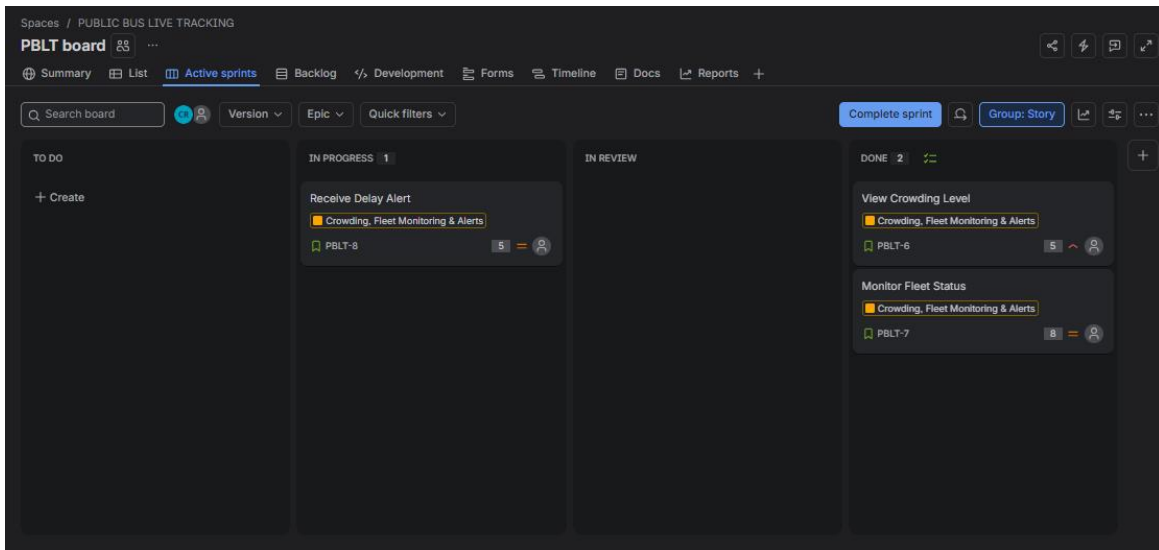


Fig 2.3 - Sprint 3 board (To Do / In Progress / In Review / Done)

Sprint 3 Burndown Chart



Fig 2.4 - Sprint 3 burndown chart (Goal: Deliver core live-tracking and ETA features)

TIME				Story Points		
Date	Issue	Event Type	Event Detail	Inc.	Dec.	Remaining
28/Aug/26 1:54 PM	Total	Sprint start				0
	PBLT-6	Scope change	Issue added to sprint	5		5
	PBLT-8	Scope change	Issue added to sprint	5		10
	PBLT-7	Scope change	Issue added to sprint	8		18
28/Aug/26 1:56 PM	PBLT-6	Burndown	Issue completed		5	13
	PBLT-7	Burndown	Issue completed		8	5
28/Aug/26 1:57 PM	PBLT-8	Burndown	Issue completed		5	0
28/Aug/26 1:58 PM	PBLT-6	Sprint ended by Chathurth R				0
	PBLT-7					0
	PBLT-8					0
	Total					0

Fig 2.5 - Sprint 3 burndown event log

Final Backlog View - Both Epics

The consolidated backlog view below shows both Epics side by side, with Epic 1 (Live Bus Tracking & ETA) at 100% completion across all three child stories.

The screenshot displays the Jira backlog view with two epics side-by-side. The left epic is 'PBLT Sprint 3' and the right epic is 'Live Bus Tracking & ETA'. The right epic is at 100% completion.

Left Epic: PBLT Sprint 3

- Version: Epic 2
- Quick filters: Clear filters
- Plan your sprint: Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.
- Create
- Backlog (3 work items): 18 0 0 0 Create sprint
- Create a whiteboard to plan your work Try
- PBLT-6 View Crowding Level To Do 5
- PBLT-7 Monitor Fleet Status To Do 8
- PBLT-8 Receive Delay Alert To Do 5
- Create

Right Epic: Live Bus Tracking & ETA

- PBLT-1
- To Do Improve Epic
- Description: Enable commuters to find buses, track their locations, and view accurate estimated arrival times.
- Child work items: 100% Done
- Work table:

Work	Pri...	Stor...	As...	Status
PBLT-2 Search Stop & View ETA	H.	5	U	Done
PBLT-3 Real-Time Bus Location	H.	5	U	Done
PBLT-4 Accurate ETA Updates	H.	8	U	Done

Linked work items

Add linked work item

Confluence content

Project plan Try template

Details

Fig 2.6 - Final backlog view with both Epics and completion status

Reflection Questions

1. Did your estimations reflect the actual effort?

Mostly yes, but not perfectly. In Sprint 1, "Search Stop & View ETA" (5 pts) and "Real-Time Bus Location" (5 pts) were completed within minutes of starting the sprint, while "Accurate ETA Updates" (8 pts) remained in To Do - suggesting the 8-point estimate correctly flagged it as the most complex story in that Epic, even though our simulated "effort" was really just moving cards rather than doing real development work. In Sprint 2, both remaining stories (5 and 8 points, 13 total) were marked Done almost immediately, which shows the simulation doesn't capture real effort duration - in an actual sprint, an 8-point story should take noticeably longer than a 5-point one. The relative sizing (8 > 5) still felt appropriate given the complexity of each story: recalculating ETAs from live GPS data and monitoring a fleet dashboard are genuinely harder than a stop search or a single crowding readout.

2. Was your backlog well-prioritized?

Yes. All the core commuter-facing stories that directly implement FR-001, FR-002, and FR-003 (Search Stop & View ETA, Real-Time Bus Location, Accurate ETA Updates, View Crowding Level) were marked High priority, since these deliver the primary value described in our Lab 1 requirements. The two stories tied to secondary/administrative functionality - Monitor Fleet Status (FR-004) and Receive Delay Alert (FR-005) - were set to Medium, since a working tracking and crowding experience for commuters delivers value even before fleet-side monitoring and proactive alerts are built. Splitting the backlog into two Epics (Live Bus Tracking & ETA, and Crowding/Fleet Monitoring/Alerts) also made it easy to prioritize Epic 1 into Sprint 1 first, since it covers the most foundational functionality the rest of the system depends on.

3. How did your simulated sprint align with your plan?

Across all three sprints, the simulation loosely tracked the plan but revealed how easy it is for a sprint's actual progress to diverge once work starts. Sprint 1 ended with a mixed state (one story Done, one In Progress, one still To Do) after a story was briefly marked Done and then reopened - a realistic occurrence that would, in an actual team, represent discovering an issue during review. Sprint 2 cleanly finished both carried-over stories. Sprint 3, covering all three Epic 2 stories together, again ended in a mixed state rather than all three finishing cleanly, which is a more realistic sprint outcome than instantly clearing the whole board. Overall, the plan (2 Epics, 6 stories, front-loading the highest-priority Epic first) held up, but the sprint-by-sprint execution was uneven - exactly the kind of variance real burndown charts are meant to expose.

4. What insights did the burndown chart give about your team's capacity?

The three charts told different stories. Sprint 1's chart showed a bump back up mid-sprint (from the reopened issue), which is a useful signal in real projects: it usually means a story was marked Done prematurely, before it actually met its acceptance criteria. Sprint 2's chart dropped straight to zero right after sprint start, which - if this were a real team rather than a simulation - would be a warning sign of under-committing during sprint planning, since work finishing instantly rarely reflects genuine effort. Sprint 3 landed between these two extremes, ending the sprint partway down the guideline line. Together, the three sprints illustrate the burndown chart's real purpose: it's less about whether the team hits zero, and more about whether the pattern (steady decline vs. spikes vs. instant drops) matches how the team actually expects to work, which helps catch estimation or scope problems early rather than only at sprint review.

